

CN question

1. What happens when you type "google.com" into your browser and press Enter?

- **DNS Resolution:** The browser checks its cache, then the OS cache, and finally queries the Domain Name System (DNS) to find the IP address associated with "google.com".
- **TCP Handshake:** The browser initiates a TCP three-way handshake with the server to establish a reliable connection.
- **TLS Handshake:** If using HTTPS, a secure TLS/SSL handshake occurs to encrypt the communication.
- **HTTP Request & Response:** The browser sends an HTTP GET request to the server, and the server responds with the HTML content.
- **Rendering:** The browser parses the HTML, fetches additional resources (CSS, JS, images), and renders the web page on your screen.

2. What is the difference between TCP and UDP?

- **TCP (Transmission Control Protocol):** Connection-oriented, highly reliable, and guarantees the delivery of data packets in the correct order. It uses error-checking and flow control. (Used for Web browsing, emails, file transfers).
- **UDP (User Datagram Protocol):** Connectionless, faster, but unreliable. It does not guarantee delivery, order, or error recovery. (Used for live video streaming, online gaming, VoIP).

3. Explain the TCP Three-Way Handshake.

The three-way handshake is the process used by TCP to set up a logical connection between a client and a server:

1. **SYN:** The client sends a Synchronize (SYN) packet to the server to initiate a connection.
2. **SYN-ACK:** The server receives the SYN packet and responds with a Synchronize-Acknowledgment (SYN-ACK) packet.
3. **ACK:** The client receives the SYN-ACK and sends a final Acknowledgment (ACK) packet back to the server. The connection is now established.

4. What are the layers of the OSI Model?

The Open Systems Interconnection (OSI) model conceptualizes network communication into 7 distinct layers:

1. **Physical:** Physical cables and hardware (Bits).
2. **Data Link:** MAC addressing and switching (Frames).
3. **Network:** IP addressing and routing (Packets).

4. **Transport:** End-to-end communication, TCP/UDP (Segments).
5. **Session:** Establishing and managing sessions.
6. **Presentation:** Data translation, encryption, and compression.
7. **Application:** Network applications and user interfaces (HTTP, FTP, SMTP).

5. What is the difference between a MAC address and an IP address?

- **MAC Address (Media Access Control):** A permanent, 48-bit physical hardware address burned into the Network Interface Card (NIC) by the manufacturer. It operates at Layer 2 (Data Link Layer) to deliver data within the same local network.
- **IP Address (Internet Protocol):** A logical address assigned by the network (either dynamically or statically). It operates at Layer 3 (Network Layer) and is used to route data across different networks over the internet.

6. What is Subnetting and why is it used?

- **Subnetting** is the practice of dividing a single, large network into multiple smaller, logical sub-networks (subnets).
- It is used to improve network performance (by reducing broadcast traffic), enhance security (by isolating network segments), and conserve the limited pool of available IPv4 addresses.

7. How does HTTPS ensure secure communication?

- HTTPS is standard HTTP transmitted over a secure TLS/SSL tunnel.
- It provides **Encryption** (so eavesdroppers cannot read the data), **Data Integrity** (so data cannot be altered during transit without detection), and **Authentication** (using digital certificates to prove the server is who it claims to be).

8. What is the difference between a Router and a Switch?

- **Switch:** Operates at the Data Link layer (Layer 2). It connects multiple devices within the *same* Local Area Network (LAN) and uses MAC addresses to forward data frames directly to the intended device.
- **Router:** Operates at the Network layer (Layer 3). It connects *different* networks together (e.g., your home LAN to the internet) and uses IP addresses to route data packets to their final destination.

9. What is DNS and how does it work?

- **DNS (Domain Name System)** is essentially the "phonebook" of the internet.
- Because computers communicate using IP addresses (like 142.250.190.46), but humans prefer memorable names (like google.com), DNS resolves or translates these human-readable domain names into the corresponding machine-readable IP addresses.

10. What is BGP (Border Gateway Protocol)?

- BGP is the core routing protocol of the internet. It is a path-vector protocol that dictates how data is routed between different autonomous systems (large networks or ISPs).
- While protocols like OSPF route traffic *within* a single organization's network, BGP is responsible for finding the most efficient paths to route traffic *between* massive networks across the globe.